

MINNEAPOLIS FLYING CLOUD, MN, USA

WMO: 726579

Lat: 44.832N

Lon: 93.471W

Elev: 907

StdP: 14.22

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 94963

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.6	-6.4	-18.3	2.1	-9.2	-14.8	2.5	-5.5	23.5	19.4	20.7	17.6	6.5	240	0.560

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	90.6	74.0	88.0	72.7	84.6	70.7	77.4	87.2	75.2	84.0	73.3	82.1	11.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.1	131.7	83.8	72.3	123.6	81.7	70.3	115.2	79.4	41.4	86.5	39.3	84.1	37.4	82.3	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.9	19.3	17.5	-18.1	96.1	5.5	2.9	-22.0	98.1	-25.3	99.8	-28.4	101.4	-32.4	103.5
			-18.4	80.6	5.4	2.3	-22.4	82.2	-25.5	83.5	-28.6	84.8	-32.5	86.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.0	16.4	19.7	32.9	47.6	59.7	69.0	74.4	71.4	64.3	49.8	35.8	21.7
	DBStd	22.53	12.72	12.17	12.42	10.21	8.58	6.60	5.34	5.32	8.33	9.36	10.66	11.89
	HDD50	4055	1042	849	547	162	15	0	0	0	4	120	439	877
	HDD65	7368	1507	1269	997	526	211	35	3	7	114	482	875	1342
	CDD50	2971	0	0	16	90	315	570	757	665	431	113	14	0
	CDD65	809	0	0	1	4	46	155	295	207	92	9	0	0
	CDH74	6854	0	0	5	58	465	1273	2662	1584	727	79	1	0
	CDH80	2194	0	0	0	14	155	393	944	458	215	15	0	0
Wind	WSAvg	8.4	8.4	8.7	9.0	9.9	9.0	8.0	7.3	7.0	8.1	8.8	8.8	8.3
Precipitation	PrecAvg	31.8	0.7	0.8	1.6	3.2	4.1	4.8	3.9	4.4	3.4	2.4	1.4	1.1
	PrecMax	40.0	1.5	1.6	3.0	6.7	9.0	9.1	6.8	8.0	7.6	5.7	4.3	2.7
	PrecMin	23.3	0.1	0.3	0.7	1.5	0.7	1.9	1.8	0.7	0.5	0.6	0.1	0.2
	PrecStd	4.9	0.4	0.4	0.6	1.3	2.0	1.9	1.7	2.3	2.2	1.5	1.1	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.6	52.0	70.9	81.2	90.8	92.4	95.3	92.6	90.5	81.9	68.2	51.8
		MCWB	38.1	46.2	58.4	62.1	68.6	74.2	77.5	75.9	72.6	65.3	55.6	47.3
	2%	DB	39.0	44.6	62.7	73.1	84.1	88.2	90.8	88.2	85.9	75.0	61.4	44.8
		MCWB	35.3	39.2	53.8	56.1	65.3	71.1	75.8	73.4	71.0	61.4	52.1	39.8
	5%	DB	36.1	39.2	56.6	69.2	79.2	84.1	88.2	84.4	81.6	69.8	55.2	39.3
		MCWB	33.5	35.5	49.1	54.0	62.9	69.5	74.2	71.6	68.0	58.8	48.0	35.9
	10%	DB	33.6	36.3	50.2	63.9	74.1	81.3	84.9	81.9	78.1	64.1	51.1	36.6
		MCWB	31.5	33.1	44.1	51.0	60.2	67.4	71.8	69.4	66.4	55.5	44.7	33.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.6	46.2	61.3	63.8	72.8	77.4	81.5	78.4	76.1	69.1	58.7	48.4
		MCDB	43.1	51.6	68.1	77.4	86.7	88.1	89.2	88.5	86.1	76.5	65.7	51.0
	2%	WB	35.8	39.5	55.2	58.9	68.2	74.2	78.4	75.9	73.0	63.6	53.5	40.3
		MCDB	38.6	43.3	61.3	70.0	79.2	84.1	87.9	84.6	81.6	70.0	60.1	43.8
	5%	WB	33.5	35.8	49.6	55.5	65.3	71.7	76.2	74.0	70.5	60.2	48.8	36.7
		MCDB	35.8	39.2	56.2	66.2	75.3	81.6	85.3	82.1	78.8	68.3	54.9	39.6
	10%	WB	31.1	33.3	44.1	52.4	62.6	69.2	74.0	71.9	68.1	56.4	45.1	34.0
		MCDB	33.1	36.0	50.0	62.5	71.7	77.9	82.8	79.6	76.1	63.6	50.1	36.6
Mean Daily Temperature Range	5% DB	MDBR	14.6	15.5	15.9	18.2	18.8	17.8	18.2	17.8	18.8	16.9	14.5	13.3
		MCDBR	15.9	16.5	22.9	26.1	26.0	21.7	20.5	20.2	22.7	24.6	21.5	16.6
		MCWBR	13.3	12.9	14.8	14.2	12.0	10.3	9.5	9.5	10.8	13.6	14.0	13.3
	5% WB	MCDBR	14.9	15.7	21.0	23.0	22.5	19.0	18.9	18.1	19.7	20.1	20.1	15.4
		MCWBR	12.8	12.4	14.5	13.9	12.0	10.9	10.2	10.3	11.1	12.9	14.1	12.8
Clear-Sky Solar Irradiance	taub	0.270	0.285	0.319	0.374	0.404	0.400	0.402	0.405	0.378	0.322	0.292	0.268	
	taud	2.421	2.362	2.390	2.300	2.270	2.327	2.338	2.328	2.366	2.520	2.509	2.440	
	Ebn at Noon	272	287	291	281	275	276	273	268	266	271	263	261	
	Edh at Noon	23	29	33	39	42	39	39	38	33	25	21	20	
All-Sky Solar Radiation	RadAvg	548	895	1190	1454	1690	1885	2003	1671	1307	832	531	387	
	RadStd	66	91	114	149	139	152	76	121	95	110	51	42	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	-0.88	N/A	N/A	N/A	N/A
(47)	Regional (60 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page